



Barker College

**2003
TRIAL
HIGHER SCHOOL
CERTIFICATE**

Mathematics Extension 1

Staff Involved:

PM THURSDAY 14 AUGUST

- CFR*
- HG*
- DOK
- RMH
- MRB
- BJR
- VAB

90 copies

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using blue or black pen
- Make sure your Barker Student Number is on ALL pages
- Board-approved calculators may be used
- A table of standard integrals is provided on page 9
- ALL necessary working should be shown in every question
- Marks may be deducted for careless or badly arranged working

Total marks – 84

- Attempt Questions 1 – 7
- All questions are of equal value

Total marks – 84

Attempt Questions 1 – 7

ALL questions are of equal value

Answer each question on a SEPARATE sheet of paper

Marks

Question 1 (12 marks) [BEGIN A NEW PAGE]

- (a) If $f(x) = x^2$ and $g(x) = -\sqrt{x}$, what is the value of $f(g(9)) - g(f(9))$? 2
- (b) $y = f(x)$ is a linear function with slope $\frac{1}{2}$
- (i) Find an expression for the inverse function of $y = f(x)$ 2
- (ii) Hence find the slope of $y = f^{-1}(x)$ 1
- (c) Find $\int \frac{2}{3\sqrt{16 - x^2}} dx$ 1
- (d) Find the coordinates of the point that divides the interval AB , where A is $(-1, 3)$ and B is $(2, 8)$, externally in the ratio of $3 : 2$. 2
- (e) If $\sin 2A = \frac{1}{2}$, what is the value of $\frac{1}{\sin A \cos A}$? 2
- (f) If $0 \leq t \leq 1$, find the Cartesian equation of the curve whose parametric equations are $y = t^2$ and $x = \sqrt{t}$ 2

Question 2 (12 marks) [BEGIN A NEW PAGE]

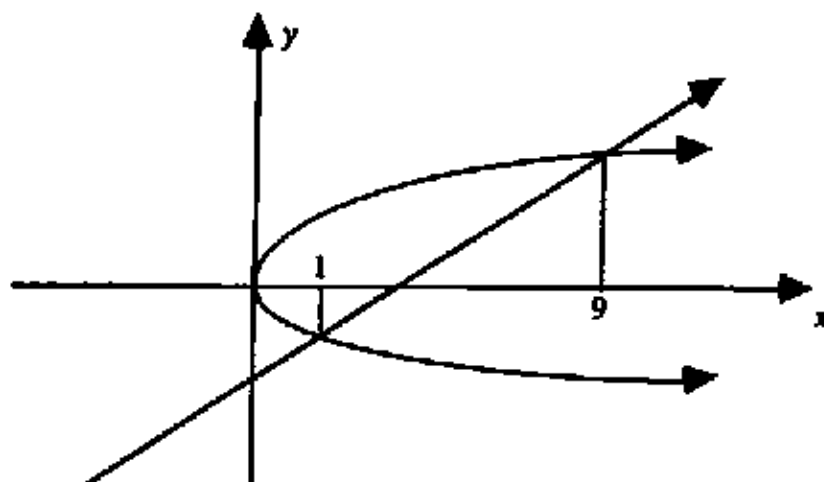
- (a) Consider the function $y = 2\sin^{-1}\frac{x}{3}$
- (i) State the domain and range of $y = f(x)$ 2
- (ii) Hence sketch the graph of $y = f(x)$ 1
- (b) From the top, C, of a vertical cliff, 200 m high, two ships P and Q are observed at sea level. A is the foot of the cliff at sea level. P is due south of A and the angle of elevation of C from P is 45° . Q is $S 50^\circ W$ of A and the angle of elevation of C from Q is 60° .
- (i) Draw a diagram showing this information. 1
- (ii) Find the distance PQ (to nearest metre). 3
- (c) Consider the curve whose equation is $y = \frac{x^2}{1 - x^2}$
- (i) Find any vertical asymptotes. 1
- (ii) Find $\lim_{x \rightarrow \pm\infty} y$ 1
- (iii) Show that the curve is an even function. 1
- (iv) Hence (without using calculus), sketch the curve, showing all main features. 2

Question 3 (12 marks) [BEGIN A NEW PAGE]

- (a) Differentiate $x \cos^{-1} x$ 2
- (b) Find $\int_0^{\pi} \sin^3 x dx$ using the result $\sin 3x = 3 \sin x - 4 \sin^3 x$ 3
- (c) A boat is attached by a rope to a jetty 2 m above the bow of the boat.
The rope is being pulled in at the rate of 1 m s^{-1} .
At what rate is the boat approaching the jetty when 3 m of rope still remains
to be pulled in? (Answer correct to 1 decimal place) 4
- (d) (i) Express $x^2 + x + 1$ in the form $(x - A)^2 + B$ where
 A, B are constants. 1
- (ii) Hence find $\int \frac{dx}{x^2 + x + 1}$ 2

Question 4 (12 marks) [BEGIN A NEW PAGE]

- (a) The curves $y^2 = 16x$ and $y = 2x - 6$ intersect at the points where $x = 1$ and $x = 9$.



Find the acute angle between the two curves at the point where $x = 1$

3

- (b) If $\tan \frac{\theta}{2} = t$ and θ is acute, express $\sqrt{\frac{1 - \cos 2\theta}{1 + \cos 2\theta}}$ in terms of t

3

- (c) Evaluate in exact form $\cos 105^\circ$

2

- (d) Solve $\sqrt{2} \cos x - \sin x = \frac{3}{2}$ for $0^\circ \leq x \leq 360^\circ$

4

Question 5 (12 marks) [BEGIN A NEW PAGE]

- (a) A body is cooling in a room of constant temperature
- 15°C
- .

At time t minutes its temperature, T , decreases according to the equation

$$\frac{dT}{dt} = -k(T - 15)$$

where k is a positive constant.The initial temperature of the body is 75°C , and it cools to 55°C after 10 minutes.

What is the temperature of the body after a further 5 minutes?

(Answer correct to 1 decimal place)

4

- (b) (i) Show that the relation $v^2 = -kx^2 + c$, where k and c are constants, is satisfied by the equation $\frac{d}{dx}\left(\frac{v^2}{2}\right) = -kx$

1

- (ii) A pendulum, P , swings so that it oscillates about its centre of motion according to the equation $\frac{d^2x}{dt^2} = \frac{-x}{9}$, where x is the distance of P from its centre of oscillation at any time t seconds.

Show that $v^2 = \frac{1}{9}(4 - x^2)$, given that its maximum displacement is 2 cm.Hence find the maximum speed of P .

4

- (c) Evaluate $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \sin x \cos^2 x \, dx$ using $u = \cos x$

3

Question 6 (12 marks) [BEGIN A NEW PAGE]

(a) Write down the value of ${}^nC_j - {}^nC_{n-j}$ 1

(b) Find the term independent of x in the expansion of $\left(x^3 + \frac{5}{x}\right)^8$ 3

(c) By considering the identity $(1+x)^{2n} = \sum_{k=0}^{2n} \binom{2n}{k} x^k$, show that

$$\sum_{k=1}^{2n} k \binom{2n}{k} = n4^n \quad 4$$

(d) What is the greatest coefficient in the expansion of $(2+3x)^{20}$? 4

Question 7 (12 marks) [BEGIN A NEW PAGE]

- (a) Given that $y = \sin x$, and using the result $\cos x = \sin\left(x + \frac{\pi}{2}\right)$, it can be shown that :

$$\begin{aligned}\frac{dy}{dx} &= \cos x \\ &= \sin\left(x + \frac{\pi}{2}\right) \\ \frac{d^2y}{dx^2} &= \cos\left(x + \frac{\pi}{2}\right) \\ &= \sin\left[\left(x + \frac{\pi}{2}\right) + \frac{\pi}{2}\right] \\ &= \sin\left[x + \frac{2\pi}{2}\right]\end{aligned}$$

Similarly :

$$\frac{d^3y}{dx^3} = \sin\left(x + \frac{3\pi}{2}\right)$$

Therefore :

$$\frac{d^n y}{dx^n} = \sin\left(x + \frac{n\pi}{2}\right)$$

Prove, by induction, that the generalisation given above,

i.e. $\frac{d^n y}{dx^n} = \sin\left(x + \frac{n\pi}{2}\right)$, is correct for all positive integers n

when $y = \sin x$

5

- (b) A particle is projected under gravity with speed $u \text{ m s}^{-1}$ and at an angle $\frac{\pi}{4}$, from a point O on horizontal ground. It strikes the ground at P , where $OP = R$.

- (i) Taking the x and y axes through O , show that the equation of the trajectory is given by $y = x - g \frac{x^2}{u^2}$

2

- (ii) Hence, or otherwise, show that $R = \frac{u^2}{g}$

2

- (iii) A ball is fired from O with velocity 30 m s^{-1} at an angle $\frac{\pi}{4}$ to the horizontal. Find the speed of the ball when it has travelled a horizontal distance of 15 m from its starting point. (Take $g = 10 \text{ m s}^{-2}$)
(Answer correct to 1 decimal place)

3

End of Paper

YEAR 12 EXT. 1 MATHS TRIAL 2003, SOLUTIONS

QUESTION 1:

$$(a) f[f(9)] - g[f(9)] = (-\sqrt{9})^2 - \sqrt{9^2} \\ = 9 + 9 = 18 \quad (1)$$

$$(b)(i) y = \frac{1}{2}x + b$$

$$\text{Inv. } x = \frac{1}{2}y + b$$

$$2x = y + c$$

$$\therefore y = 2x - c$$

$$(ii) m_{\text{inv.}} = 2$$

$$(c) \int \frac{2}{3\sqrt{4-x^2}} dx = \frac{2}{3} \int \frac{1}{\sqrt{4-x^2}} dx \\ = \frac{2}{3} \sin^{-1} \frac{x}{2} + C \quad (1)$$

$$(d) A(-1, 3) \quad B(2, 8)$$

$$\text{Point} = \left(\frac{2(-1) + 3(2)}{-1+2}, \frac{2(3) + 3(8)}{-1+2} \right) \\ = (8, 18) \quad (1)$$

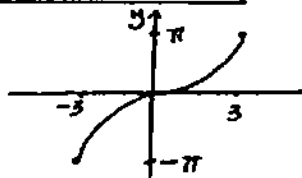
$$(e) \frac{1}{\sin A \cos A} = \frac{1}{\frac{1}{2} \sin 2A} \\ = \frac{1}{\frac{1}{2} \times \frac{1}{2}} \\ = 4 \quad (1)$$

$$(f) y = t^2, \quad x = \sqrt{t} \\ \therefore t = x^2 \\ \therefore y = (x^2)^2 = x^4 \quad (1)$$

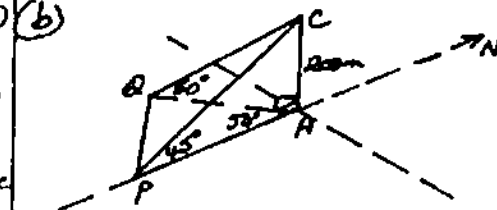
QUESTION 2:

$$(a)(i) y = 2 \sin^{-1} \frac{x}{3} \\ \text{Dom. } -1 \leq \frac{x}{3} \leq 1 \\ \therefore -3 \leq x \leq 3 \quad (1) \\ \text{Range: } -\frac{\pi}{2} \leq \sin^{-1} \frac{x}{3} \leq \frac{\pi}{2} \\ \therefore -\pi \leq y \leq \pi \quad (1)$$

(ii)



(b)



$$QA = \frac{200}{\tan 60^\circ} \\ = 115.47$$



$$PA = 200 \quad (\text{opp. to } \angle A)$$



$$QP^2 = (115.47)^2 + 200^2 - 2 \times 115.47 \times 200 \cos 50^\circ$$

$$= 23644.24678$$

$$\therefore QP \approx 154 \text{ m (nearest m)} \quad (1)$$

$$(c) y = \frac{x^2}{1-x^2}$$

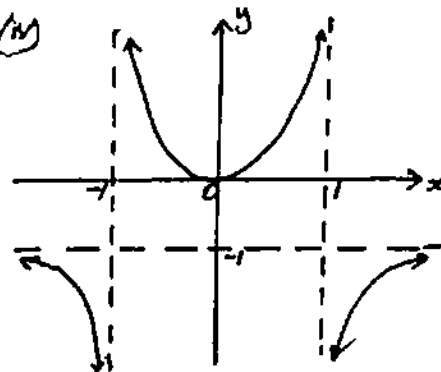
$$(i) \text{ Vert. asym. } x \neq \pm 1 \quad (1)$$

$$(ii) \lim_{x \rightarrow \pm \infty} \frac{x^2}{1-x^2} = \lim_{x \rightarrow \pm \infty} \frac{1}{\frac{1}{x^2} - 1} \\ = -1 \quad (1)$$

$$(iii) f(x) = \frac{x^2}{1-x^2} \\ f(-x) = \frac{(-x)^2}{1-(-x)^2} \\ = \frac{x^2}{1-x^2} = f(x)$$

$$\therefore \text{Even} \quad (1)$$

2 (a/v)



$$\frac{dl}{dt} = \frac{dl}{dx} \times \frac{dx}{dt}$$

$$-1 = \frac{\sqrt{5}}{\sqrt{5+4}} \frac{dx}{dt}$$

$$\therefore \frac{dx}{dt} = -\frac{3}{\sqrt{5}}$$

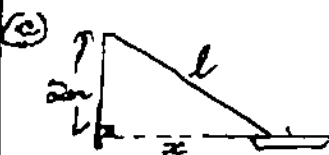
$$\therefore \text{Boat app. jettty at } 1.34 \text{ m/s} \quad (1)$$

$$(d) x^2 + x + 1 = x^2 + x + \left(\frac{1}{2}\right)^2 + 1 - \frac{1}{4} \\ = \left(x + \frac{1}{2}\right)^2 + \frac{3}{4}$$

QUESTION 3:

$$(a) \text{ Let } y = x \cos^{-1} x \quad \text{Drule} \\ \frac{dy}{dx} = \cos^{-1} x + x \cdot \frac{-1}{\sqrt{1-x^2}} \\ = \cos^{-1} x - \frac{x}{\sqrt{1-x^2}}$$

$$(b) \sin 3x = 3 \sin x - 4 \sin^3 x \\ \therefore \sin^3 x = \frac{1}{4} (3 \sin x - \sin 3x) \\ \int_0^{\pi} \sin^3 x dx = \frac{1}{4} \int_0^{\pi} (3 \sin x - \sin 3x) dx \\ = \frac{1}{4} \left[-3 \cos x + \frac{1}{3} \cos 3x \right]_0^{\pi} \\ = \frac{1}{4} \left\{ (-3 \cos \pi + \frac{1}{3} \cos 3\pi) - (-3 \cos 0 + \frac{1}{3} \cos 0) \right\} \\ = \frac{1}{4} \left\{ (3 - \frac{1}{3}) - (-3 + \frac{1}{3}) \right\} \\ = \frac{4}{3} \quad (1)$$



$$l = \sqrt{x^2 + 4}, \quad \frac{dl}{dx} = \frac{x}{\sqrt{x^2 + 4}}, \quad l = 3, \quad x = \sqrt{5} \\ \frac{dl}{dx} = \frac{\sqrt{5}}{\sqrt{5+4}} = \frac{\sqrt{5}}{3} \quad (1)$$

QUESTION 4:

$$(a) \text{ At } x=1, y = -4\sqrt{x} \\ \frac{dy}{dx} = -2x^{-\frac{1}{2}} \\ \therefore m_1 = -2 \quad (\text{at } x=1) \\ y = 2x - 6 \\ m_2 = 2 \\ \tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \\ = \left| \frac{-2 - 2}{1 + (-2)(2)} \right| \\ = \left| \frac{-4}{-3} \right| = \frac{4}{3}$$

$$\therefore \theta \approx 53.8^\circ \quad (1)$$

$$9(b) \frac{1 - \cos 2\theta}{1 + \cos 2\theta} = \frac{1 - (1 - 2\sin^2 \theta)}{1 + (1 - 2\sin^2 \theta)} \quad ①$$

$$= \frac{2\sin^2 \theta}{2 - 2\sin^2 \theta}$$

$$= \frac{2\sin^2 \theta}{2\cos^2 \theta}$$

$$= \tan^2 \theta$$

$$= \frac{2t}{1-t^2} \quad ①$$

$$(c) \cos 105^\circ = \cos(60^\circ + 45^\circ) \quad ①$$

$$= \cos 60^\circ \cos 45^\circ - \sin 60^\circ \sin 45^\circ$$

$$= \frac{1}{2} \times \frac{1}{\sqrt{2}} - \frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{2}}$$

$$= \frac{1 - \sqrt{3}}{2\sqrt{2}} \quad ①$$

$$(d) \sqrt{2} \cos \alpha - \sin \alpha = R \cos(\alpha + \theta) \quad ①$$

$$R = \sqrt{(\sqrt{2})^2 + 1} = \sqrt{3}$$

$$\cos \alpha = \frac{\sqrt{3}}{\sqrt{3}} \quad ①$$

$$\therefore \alpha = 35^\circ 16'$$

$$\therefore \sqrt{3} \cos(\alpha + 35^\circ 16') = \frac{3}{2} \quad ①$$

$$\cos(\alpha + 35^\circ 16') = \frac{\sqrt{3}}{2}$$

$$\therefore \alpha + 35^\circ 16' = 30^\circ, 330^\circ, 390^\circ, \dots$$

$$\therefore \alpha = 294^\circ 44' \text{ or } 354^\circ 44' \quad ①$$

QUESTION 5:

$$(a) T = 15 + Ae^{-kt}$$

$$t=0 \therefore 75 = 15 + Ae^0$$

$$T=75 \therefore A=60 \quad ①$$

$$t=10 \therefore 35 = 15 + 60e^{-10k}$$

$$T=35 \therefore \frac{2}{3} = e^{-10k} \quad ①$$

$$k = -\frac{1}{10} \ln \frac{2}{3}$$

$$\text{(or } \frac{1}{10} \ln \frac{3}{2})$$

$$T = 15 + 60e^{\frac{1}{10} \ln \frac{3}{2}}$$

$$t=15, T = 15 + 60e^{\frac{3}{2} \ln \frac{3}{2}} \quad ①$$

$$= 47.659 \approx 47.7^\circ \quad ①$$

$$(b)(i) v^2 = -kx^2 + C$$

$$\frac{1}{2}v^2 = -\frac{k}{2}x^2 + C_1 \quad ①$$

$$\therefore \frac{d}{dx}\left(\frac{v^2}{2}\right) = -kx$$

$$(ii) \frac{d^2x}{dt^2} = -\frac{x}{9}$$

$$\text{i.e. } \frac{d}{dt}\left(\frac{dx}{dt}\right) = -\frac{x}{9}$$

$$\therefore \frac{1}{2}v^2 = -\frac{x^2}{18} + C \quad ①$$

$$v=0 \quad \therefore 0 = -\frac{4}{18} + C \quad ①$$

$$\therefore C = \frac{2}{9}$$

$$\therefore \frac{1}{2}v^2 = -\frac{x^2}{18} + \frac{2}{9}$$

$$\therefore v^2 = -\frac{x^2}{9} + \frac{4}{9}$$

$$= \frac{1}{9}(4 - x^2)$$

$$\text{Max } v \text{ when } x=0 \therefore v^2 = \frac{4}{9}$$

$$v = \pm \frac{2}{3}$$

$$\therefore \text{Max speed } \frac{2}{3} \text{ cm s}^{-1} \quad ①$$

$$(c) \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \sin x \cos^2 x dx$$

$$\begin{cases} u = \cos x \\ du = -\sin x dx \\ x = \frac{\pi}{3}, u = \frac{1}{2} \\ x = \frac{\pi}{2}, u = 0 \end{cases} \quad ①$$

$$= \int_{\frac{1}{2}}^0 -u^2 du$$

$$= \left[-\frac{u^3}{3}\right]_{\frac{1}{2}}^0 \quad ①$$

$$= 0 - \left(-\frac{1}{24}\right) = \frac{1}{24} \quad ①$$

QUESTION 6:

$$(a) {}^nC_j - {}^nC_{n-j} = 0 \quad ①$$

$$(b) \left(x^3 + \frac{5}{x}\right)^8$$

$$T_{r+1} = {}^8C_r x^{3(8-r)} \left(\frac{5}{x}\right)^r \quad ①$$

$$= {}^8C_r 5^r x^{24-4r} \quad ①$$

$$\therefore 24 - 4r = 0 \text{ for } T \text{ indep. of } x$$

$$\therefore r = 6$$

$$\therefore T_7 = {}^8C_6 5^6 \quad ①$$

$$= 437500$$

$$6.(c) (1+x)^{2n} = \sum_{k=0}^{2n} \binom{2n}{k} x^k$$

$$\text{Differentiating both sides wrt } x:$$

$$2n(1+x)^{2n-1} = \sum_{k=1}^{2n} k \binom{2n}{k} x^{k-1} \quad ② \text{ 1-nd side}$$

$$\text{Let } x=1, 2n \times 2^{2n-1} = \sum_{k=1}^{2n} k \binom{2n}{k} \quad ①$$

$$\therefore n 2^{2n} = \sum_{k=1}^{2n} k \binom{2n}{k}$$

$$\therefore \sum_{k=1}^{2n} k \binom{2n}{k} = n 4^n \quad ①$$

$$(d) (2+3x)^{30}$$

$$\frac{\text{coeff. } T_{r+1}}{\text{coeff. } T_r} = \frac{n-r+1}{r} \times \frac{b}{a}$$

$$= \frac{31-r}{r} \times \frac{3}{2} \quad ①$$

$$\text{If } T_{r+1} > T_r \text{ then:}$$

$$93 - 3r > 2r$$

$$5r < 93$$

$$r < 18\frac{3}{5} \quad ② \text{ method}$$

$$\therefore r = 15, 17, 16, \dots$$

$$\therefore T_{19} > T_{18} > T_{17} > \dots$$

$$\therefore T_{19} \text{ coeff. is greatest.}$$

$$2 \text{ coeff. } T_{19} = \binom{30}{18} 2^{12} 3^{18}$$

$$= \binom{30}{18} 2^{12} 3^{18} \quad ①$$

$$\text{If } T_{r+1} < T_r \text{ then:}$$

$$93 - 3r < 2r$$

$$5r > 93$$

$$r > 18\frac{3}{5}$$

$$\therefore r = 19, 20, \dots$$

$$\therefore T_{20} < T_{19}, T_{21} < T_{20}, \dots$$

QUESTION 7:

$$(a) \text{STEP 1: Prove true for } n=1$$

$$y = \sin x$$

$$\frac{dy}{dx} = \cos x = \sin\left(x + \frac{\pi}{2}\right) \quad ①$$

$$\therefore \text{True for } n=1$$

$$\text{STEP 2: Assume true for } n=k$$

$$\text{i.e. assume } \frac{d^k y}{dx^k} = \sin\left(x + \frac{k\pi}{2}\right) \quad ①$$

7. (a) cont. STEP 3: Prove true for $n=k+1$

i.e. Prove $\frac{d^{k+1}y}{dx^{k+1}} = \sin\left[x + \frac{(k+1)\pi}{2}\right]$

Now $\frac{d^{k+1}y}{dx^{k+1}} = \frac{d}{dx}\left(\frac{d^k y}{dx^k}\right)$

$= \frac{d}{dx} \sin\left(x + \frac{k\pi}{2}\right)$ from assump.

$= \cos\left(x + \frac{k\pi}{2}\right)$

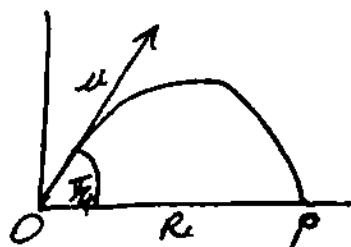
$= \sin\left(x + \frac{k\pi}{2} + \frac{\pi}{2}\right)$

$= \sin\left[x + \frac{(k+1)\pi}{2}\right]$

③ including 1 for conclusion if it follows from proof

Hence if true for $n=k$, Then true for $n=k+1$.
But true for $n=1 \therefore$ true for $n=2, n=3$ and all positive integers n .

(b) (i)



Initially: $\dot{x} = u \cos \frac{\pi}{4} = \frac{u}{\sqrt{2}}$, $\dot{y} = u \sin \frac{\pi}{4} = \frac{u}{\sqrt{2}}$
 $x=0$, $y=0$

Horiz.

$\ddot{x} = 0$

$\dot{x} = C$

$= \frac{u}{\sqrt{2}}$

$x = \frac{u}{\sqrt{2}} t + C_1$

$t=0$, $x=0 \therefore C=0$

$\therefore x = \frac{u}{\sqrt{2}} t$ ①

Vert.

$\ddot{y} = -g$

$\dot{y} = -gt + K$

$t=0$, $\dot{y} = \frac{u}{\sqrt{2}} \therefore K = \frac{u}{\sqrt{2}}$

$\therefore \dot{y} = -gt + \frac{u}{\sqrt{2}}$

$y = -\frac{gt^2}{2} + \frac{u}{\sqrt{2}} t + K_1$

$t=0$, $y=0 \therefore K_1=0 \therefore y = -\frac{gt^2}{2} + \frac{u}{\sqrt{2}} t$ ②

Now: $t = \frac{\sqrt{2}x}{u}$

$\therefore y = -\frac{g}{2} \left(\frac{\sqrt{2}x}{u}\right)^2 + \frac{u}{\sqrt{2}} \times \frac{\sqrt{2}x}{u}$

$= x - g \frac{x^2}{u^2}$

(ii) At P, $y=0 \therefore 0 = x\left(1 - \frac{gx}{u^2}\right)$

$\therefore x=0$ or $\frac{u^2 - gx}{u^2}$

$\therefore OP = R = \frac{u^2}{g}$

$R = \frac{v^2 \sin 2\alpha}{g}$

$= \frac{u^2 \sin \frac{\pi}{2}}{g} = \frac{u^2}{g}$

OR
① derived

(iii) From ① $15 = \frac{30}{\sqrt{2}} t$

$\therefore t = \frac{15\sqrt{2}}{30}$

$\dot{x} = \frac{30}{\sqrt{2}} = 15\sqrt{2}$

$\dot{y} = -10t + \frac{30}{\sqrt{2}}$

$= -5\sqrt{2} + 15\sqrt{2} = 10\sqrt{2}$

$\therefore \text{Speed} = \sqrt{(15\sqrt{2})^2 + (10\sqrt{2})^2}$
 $= \sqrt{450 + 200}$
 $= \sqrt{650}$
 $\approx 25.5 \text{ ms}^{-1}$